Lab Task 1:

Write the program for the Bubble Sort algorithm and explain its time complexity, advantages, and disadvantages.

Code:

```
#include <iostream>
using namespace std;

int main()
{
    int arr[10];

    for (int i = 0; i < 10; i++) {
        cout << "arr[" << i << "]= ";
        cin >> arr[i];
    }

    for (int i = 0; i < 10 - 1; i++) {
        for (int j = 0; j < 10 - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                int temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
    cout << "\nSorted Array: ";
    for (int i = 0; i < 10; i++) {
        cout << arr[i] << " ";
    }
}
```

Output:

```
arr[0]= 55
arr[1]= 54
arr[2]= 685
arr[3]= 2
arr[4]= 1
arr[5]= 322
arr[6]= 2
arr[7]= 6
arr[8]= 10
arr[9]= 655

Sorted Array: 1 2 2 6 10 54 55 322 655 685
```

Lab Task 2:

Write the program for the Selection Sort algorithm and explain its time complexity, advantages, and disadvantages.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[10];

    for (int i = 0; i < 10; i++) {
        cout << "arr[" << i << "]= ";
        cin >> arr[i];
    }

    for (int i = 0; i < 10 - 1; i++) {
        int minIndex = i;
        for (int j = i + 1; j < 10; j++) {
            if (arr[j] < arr[minIndex]) {
                minIndex = j;
            }
        }
        int temp = arr[i];
        arr[i] = arr[minIndex];
        arr[minIndex] = temp;
    }

    cout << "\nSorted Array: ";
    for (int i = 0; i < 10; i++) {
        cout << arr[i] << " ";
    }
}
```

Output:

```
arr[0]= 84
arr[1]= 35
arr[2]= 15
arr[3]= 65
arr[4]= 356
arr[5]= 156
arr[6]= 22
arr[7]= 753
arr[8]= 553
arr[9]= 1551

Sorted Array: 15 22 35 65 84 156 356 553 753 1551
```

Lab Task 3:

Write the program for the Insertion Sort algorithm and explain its time complexity, advantages, and disadvantages.

Code:

```
#include <iostream>
using namespace std;

int main() {
    int arr[10];

    for (int i = 0; i < 10; i++) {
        cout << "arr[" << i << "]= ";
        cin >> arr[i];
    }

    for (int i = 1; i < 10; i++) {
        int key = arr[i];
        int j = i - 1;

        while (j >= 0 && arr[j] > key) {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }

    cout << "\nSorted Array: ";
    for (int i = 0; i < 10; i++) {
        cout << arr[i] << " ";
    }
}
```

Output:

```
arr[0]= 54
arr[1]= 684
arr[2]= 57
arr[3]= 6582
arr[4]= 15
arr[5]= 5
arr[6]= 8
arr[7]= 54
arr[8]= 65
arr[9]= 6

Sorted Array: 5 6 8 15 54 54 57 65 684 6582
```